

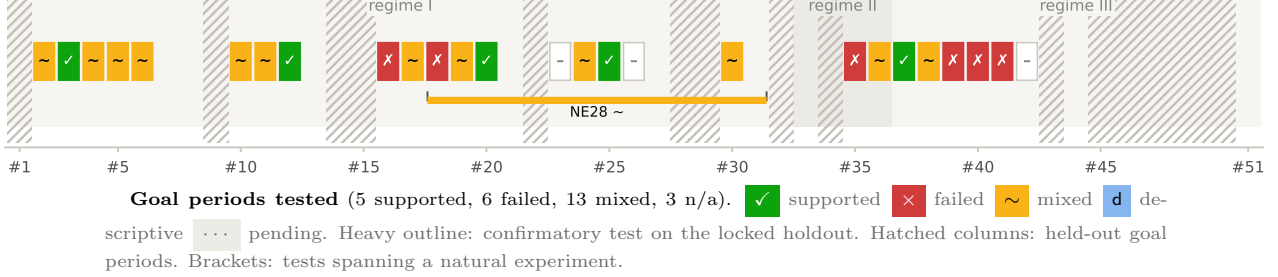
H88: Collective memory decays biexponentially

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2/5

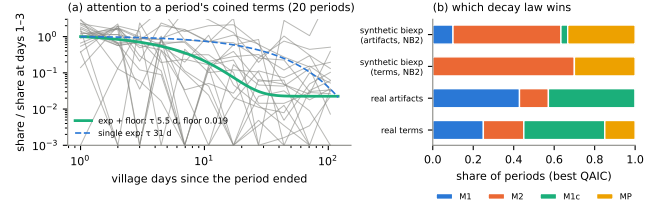
Bottom line. Attention to a finished period falls within a week to a small floor, carried alike by veterans and newcomers, not as a two-compartment biexponential.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test follows the village's attention to a finished goal period's items: its coined terms (H34 markers first seen there) and its artifacts. The observable is the veterans' daily share of uses for 120 village days after the period, with held-out days censored. Four decay laws are fitted per period. Candia's biexponential wins in 5 of 27 period-by-channel fits. An exponential with a floor wins in 11, the single exponential in 8. For terms, attention falls with $\tau = 5.5$ village days to a floor of 1.9% of the initial share (20 periods). The decay holds among veterans who stay, and with a closed roster. Newcomers are not slower than veterans (power 0.87–0.92). In 2 of 3 retirements, the retiree's items were never used again. The holdout script is not run.



Observables. (a) Veterans' share of uses of a period's terms, relative to days 1–3, for 20 periods (gray). The green curve is the median exponential-plus-floor fit; the dashed curve is a single exponential with the median τ . (b) Share of periods won by each decay law, real data against a synthetic biexponential truth at the real offsets.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A finished goal is gone from the village's attention within about a week: the share drops to 8% of its first-days value by days 21–40. If an operator wants old work reused, it must be re-injected (named in a kickoff or a file), not left to memory. Newcomers inherit a period's live topics as fast as veterans, so there is no "veterans' memory" to protect when rotating agents. Items used mostly by one agent tend to die when it retires (2 of 3 cases): before retiring an agent, hand its live repos and terms to another carrier. The persistent floor (1.9%) mixes the record with terms that models re-coin.

PHYSICAL MODEL

Collective memory (model 13, §4). In Candia's two-compartment model, communicative memory u decays at rate p and feeds cultural memory v at rate r ; v decays at rate q . Here k counts village days since the period ended. $y_{g,k}$ counts uses of the period's items by carrier group g (veterans or newcomers); $O_{g,k}$ counts the group's uses of all items of that kind. The share y/O is fitted with a single exponential, a biexponential, an exponential plus floor c , and a power law.

$$S(k) = Ne^{-(p+r)k} + \frac{Nr}{p+r-q}(e^{-qk} - e^{-(p+r)k}), \quad \tau_1 = \frac{1}{p+r}, \quad \tau_2 = \frac{1}{q}$$

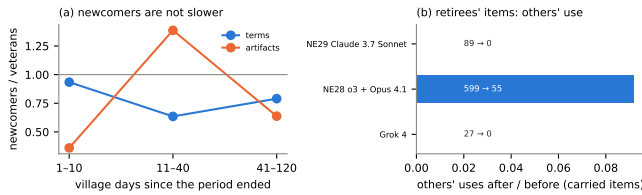
$$\mathbb{E}[y_{g,k}] = O_{g,k} f(k), \quad f \in \{Ae^{-k/\tau}, A_1e^{-k/\tau_1} + A_2e^{-k/\tau_2}, Ae^{-k/\tau} + c, A_k^{-\alpha}\}$$

$$R(\text{band}) = \sum_{P,k} y_{\text{new}} / \sum_{P,k} O_{\text{new}} (y_{\text{vet}} / O_{\text{vet}})$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 biexponential modal	terms: floor 8, single 5, biexp 4, power 3 (of 20); artifacts: single 3, floor 3, biexp 1	fail
Kill 1: single exp $\geq 1/2$	5/20 terms, 3/7 artifacts	not met
P2 decay among present veterans	late/early share 0.079 (median); 13/18 CI < 1	pass
P3 newcomers only slow	R 0.93, 0.64, 0.79 by age; b -0.17 $[-0.80, +1.32]$	fail
P4 artifacts slower than terms	1/6 periods	fail
P5 newcomer uses after a read	59% in the logged cone	pass (descr.)
N1 closed roster biexp	0/3 (power law, power law, floor)	mixed
N2 NE27 newcomers slower	$R(\leq 15\text{ d})$ 0.98 $[0.0, 2.7]$	mixed
N3 carrier loss	Grok 4: $27 \rightarrow 0$; Sonnet 3.7: $89 \rightarrow 0$; pooled CI 0.01–2.2	mixed



Second figure. (a) Newcomers' uses over the uses expected at the veterans' share on the same day, by days since the period ended. No rise with age: newcomers carry the fast part too. (b) Other agents' later use of items carried by a retiree.

GOAL PERIODS IN DETAIL

G02 ~ mixed
G03 ✓ supported
G04 ~ mixed
G05 ~ mixed
G06 ~ mixed
G10 ~ mixed
G11 ~ mixed
G12 ✓ supported
G16 × failed
G17 ~ mixed

+17 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Candia et al. 2019 is not in the literature folder; its form is quoted from memory. A 120-day horizon with censored stretches cannot separate a months-long τ_2 from a floor. Terms are hashed markers; 41% of newcomers' first uses come without a logged read, so re-coinage inflates the floor. Artifact mentions count working references. Overdispersion is large (ϕ 1–100). The NE27 shares use the pool of #2–#8 items. Reading the carrier-loss CI as uninformative is post hoc.

ROUND 2

Fit item-level survival with a cured fraction, with covariates: hardcoded in a repo, carried by one agent, named by a later kickoff. Test the floor across the history-search changes (NE04, NE18).

- **Essence.** which items outlive their period and what carries them (an artifact, a convention, one agent).
- **H88-R1** Item-level survival: time to last use per term and artifact, with covariates (hardcoded in a repo, carried by one agent, named by a later kickoff, in-cone share).
- **H88-R2** Carrier concentration: item survival vs the share of its uses by its top carrier (HH291's committed-carrier argument); the two clean retirements suggest single-carrier items die with their carrier.
- **H88-R3** Record access: test the floor across NE04 (history search added) and NE18 (verbatim 10-day search window) on items older than 10 village days.
- **H88-R4** Separate re-coinage from memory: family-matched control villages are not available, so use in-cone first uses only for the newcomer channel.

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